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Paper Title

LoRA-Based Resource-Efficient LLM Adaptation for RTL Hardware Trojan Detection

Abstract

As the use of third-party intellectual property (IP) cores and the complexity of hardware design workflows continue to grow, Hardware Trojan detection at the Register Transfer Level (RTL) is becoming increasingly critical. Many existing hardware deep learning (DL) and transformer-based approaches to RTL security analysis have high computational resource demands and do not support reproducibility, particularly in academic settings with limited resources. In this work, we present an engineered prototype, lightweight and reproducible, for RTL Trojan analysis using parameter-efficient large language model (LLM) adaptation. Our proposed approach employs LoRA-based fine-tuning of the Qwen2.5-1.5B-Instruct model for Trojan detection and localization in RTL designs. A custom synthetic RTL Trojan dataset of clean and Trojan-only Verilog samples was developed using controlled trigger and payload insertion. To prevent shortcut learning, we utilized both lightweight structural obfuscation and prompt normalization methods at the preprocessing stage. Evaluation was performed on a randomly stratified test split of the dataset and included both classification and localization analyses using Trigger Localization Coverage (TLC) and Payload Localization Coverage (PLC). The entire implementation was executed on a Google Colab Tesla T4 GPU using low-resource LoRA adaptation. The framework achieved 98.89\% primary metric accuracy, 98.88\% F1-score, 98.89\% TLC, and 98.89\% PLC metrics while also maintaining a computationally-efficient training setup suitable for reproducible experimentation.

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